FIITJEE

ALL INDIA TEST SERIES

PART TEST – II

JEE (Main)-2024

TEST DATE: 03-12-2023

ANSWERS, HINTS & SOLUTIONS

Physics

PART – A

SECTION – A

1. C

Sol. $K_1 A \tan 30° = K_2 A \tan 45° = K_3 A \tan 60°$

$\Rightarrow K_1 \tan 30° = K_2 \tan 45° = K_3 \tan 60°$

$$\Rightarrow \frac{K_1}{3\sqrt{3}} = \frac{K_2}{3} = \frac{K_3}{\sqrt{3}}$$

$$\Rightarrow \frac{K_1}{3} = \frac{K_2}{\sqrt{3}} = \frac{K_3}{1}$$

2. C

Sol. $$\frac{\mu_0 I_1}{4\pi(a+x)} = \frac{\mu_0 I_2}{4\pi(a-x)}$$

$$\Rightarrow \frac{a-x}{a+x} = \frac{I_2}{I_1}$$

$\Rightarrow I_1 a - I_1 x = I_2 a + I_2 x$

$$\Rightarrow \quad x = \left(\frac{I_1 - I_2}{I_1 + I_2}\right) a$$

3. C

Sol. The current at time t is given by

$i = i_0\left(1 - e^{-t/\tau}\right)$

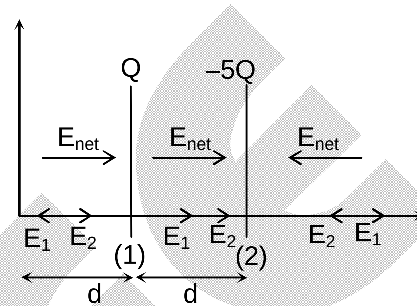
Here, $i_0 = \frac{E}{R}$ and $\tau = \frac{L}{R}$

$$\therefore\ q = \int_0^{\tau} i\,dt = \int_0^{\tau} i_0\left(1 - e^{-t/\tau}\right)dt$$

$$= \frac{i_0\tau}{e} = \frac{\left(\frac{E}{R}\right)\left(\frac{L}{R}\right)}{e} = \frac{EL}{eR^2}$$

4. A

Sol. Potential decrease in the direction of E.



$\therefore$ From origin potential decrease from V_0 to certain value

$E = -\frac{dV}{dx}$

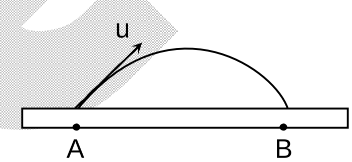
$\therefore$ Slope of graph V-x is equal to E

E is high in between the plate

$\therefore$ Slope between plate is maximum.

5. B

Sol. Field near metal surface $E = \frac{\sigma}{\varepsilon_0}$



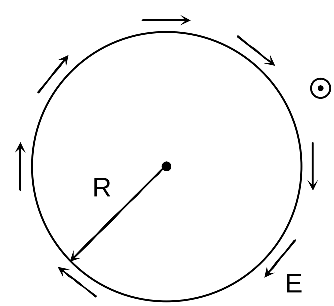
Force on electron $eE = \frac{e\sigma}{\varepsilon_0}$

Acceleration of electron $a = \frac{e\sigma}{m\varepsilon_0}$

Maximum range $\frac{u^2}{a} = \frac{u^2 m\varepsilon_0}{\sigma e}$

6. D

Sol. Due to the time varying magnetic field, an induced electric field E is set up, so the force on the ring is



F = qE

If f be the frictional force between the ring and the table, then the ring starts rotating when

F = f

$\Rightarrow qE = \mu mg$ …(1)

Also, $\oint \vec{E}\cdot d\vec{\ell} = \left|\frac{d\phi}{dt}\right|$

$\Rightarrow\ E(2\pi R) = A\left(\frac{dB}{dt}\right)$

$\Rightarrow\ E(2\pi R) = (\pi R^2)(16t)$ $\left\{\because \frac{dB}{dt} = 16t\right\}$

$\Rightarrow E = 8Rt$ …(2)

Substituting (2) in (1), we get

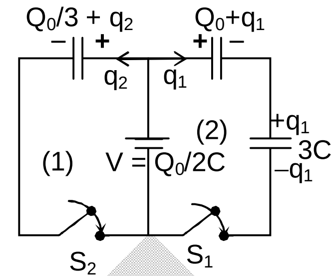
$q(8Rt) = \mu mg$

$\Rightarrow\ \mu = \left(\frac{8qR}{mg}\right)t$

$\Rightarrow\ \mu\big|_{t=3} \frac{24qR}{mg}$

7. A

Sol. In loop (1)



$$V - \left(\frac{\frac{Q_0}{3} + q_2}{C}\right) = 0$$

$$\Rightarrow q_2 = \frac{Q_0}{6}$$

In loop (2)

$$V - \left(\frac{Q_0 + q_1}{2C}\right) - \frac{q_1}{3C} = 0$$

$$\Rightarrow q_1 = 0$$

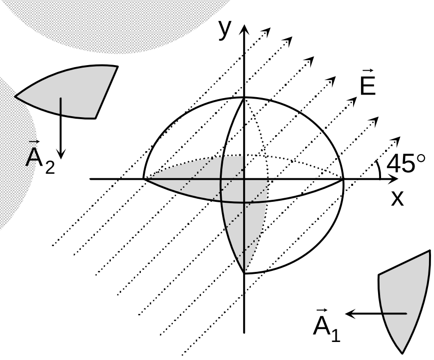
8. A

Sol. Coefficient of performance = $\dfrac{\text{heat extracted}}{\text{work done}}$

$$W = \frac{2000}{4} = 500 \text{ cal} = 2100\,\text{J}$$

9. C

Sol. $\phi_{plane} + \phi_{curve} = 0$ or $\phi_{plane} = -\phi_{curve}$



$$\vec{A}_1 = -\frac{\pi R^2}{2}\hat{i},\ \vec{A}_2 = -\frac{\pi R^2}{2}\hat{j}$$

$$\vec{E} = E\cos 45°\hat{i} + E\cos 45°\hat{j}$$

$$= \frac{E}{\sqrt{2}}\hat{i} + \frac{E}{\sqrt{2}}\hat{j}$$

$$\phi = \vec{E}\cdot(\vec{A}_1 + \vec{A}_2)$$

$$= \frac{-E}{\sqrt{2}}\frac{\pi R^2}{2} - \frac{E}{\sqrt{2}}\frac{\pi R^2}{2} = \frac{-\pi R^2 E}{\sqrt{2}}$$

This is the flux entering. So flux is $\dfrac{\pi R^2 E}{\sqrt{2}}$

10. B

Sol. Potential drops across C_1 and C_2 are 6V and 4V, respectively. Since they are in series, same charge flows through them and

$$\frac{C_1}{C_2} = \frac{V_2}{V_1} = \frac{2}{3}$$

11. A

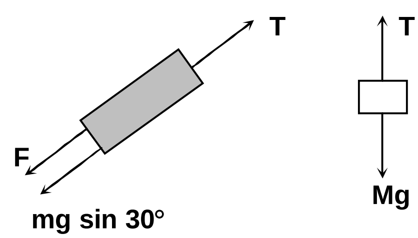
Sol. For equilibrium, we have

T = Mg

and T = F + sin30°

or $Mg = F + \dfrac{mg}{2}$

or $M = \dfrac{m}{2} + \dfrac{F}{g} = \dfrac{m}{2} + \dfrac{E^2\varepsilon_0 A(k-1)}{2\ell g d}$

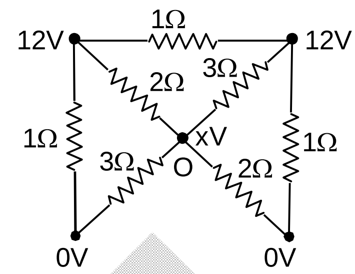


12. D

Sol. The circuit can be redrawn as

At junction O, let potential of this point is x.

$$\frac{(12-x)}{2}+\frac{(12-x)}{3}=\frac{(x-0)}{3}+\frac{(x-0)}{2} \text{ or } x = 6V$$



13. C

Sol. $R_{AB}=16\Omega$, $V_{AB}=\frac{R_{AB}\varepsilon_0}{R+R_{AB}}=\frac{16\times12}{8+16}=8V$

Current in the loop of ε_1 and ε_2:

$$I=\frac{\varepsilon_2+\varepsilon_1}{r_1+r_2}=\frac{4+2}{2+6}=\frac{3}{4}A$$

$$\varepsilon_1-Ir=V_{AN}=\frac{AN}{AB}(V_{AB})$$

or $2-\frac{3}{4}\times2=\frac{AN}{4}\times8$

or $AN=\frac{1}{4}m=25$ cm

14. B

Sol. Work done

$W = -MB(\cos\theta_2 - \cos\theta_1)$

i.e., $W_{90}=-MB(\cos90°-\cos0°)=MB$

and $W_{60}=-MB(\cos60°-\cos0°)=\frac{MB}{2}$

It is given that,

$W_{90}=nW_{60}$

$\therefore n=\frac{W_{90}}{W_{60}}=\frac{MB}{(MB/2)}=2$

15. C

Sol. $\chi_m \propto \frac{1}{T}$

$\therefore \frac{(\chi_m)_{300K}}{(\chi_m)_{350K}}=\frac{350}{300}$

or $(\chi_m)_{300K}=\frac{350}{300}\times2.8\times10^{-4}=3.267\times10^{-4}$

16. B

Sol. $\frac{I_P}{I_S}=\frac{N_S}{N_P}$

$I_P=\frac{N_S}{N_P}\times I_S=\frac{25}{1}\times2=50$ A

17. B

Sol. $PV^{4/3} = \text{constant}$

$\Rightarrow P^3V^4 = \text{constant}$

$\Rightarrow 3\ell nP + 4\ell nV = \text{constant}$

$\Rightarrow 4\frac{dV}{V} = -\frac{3dP}{P}$

$\Rightarrow 4PdV + 3VdP = 0$

$\Rightarrow 3(PdV + VdP) + PdV = 0$

For adiabatic process,

$dQ = dU + dW = 0$

$\Rightarrow dU = -PdV$ …(i)

$\Rightarrow -PdV = 3(PdV + VdP)$ …(ii)

From equation (i) and (ii)

$\Rightarrow dU = 3(PdV + VdP)$

$\Rightarrow \int dU = 3\int (PdV + VdP)$

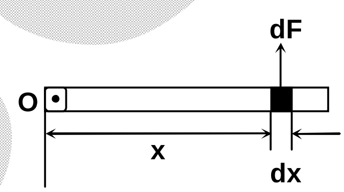
$\Rightarrow U = 3\int d(PV)$

$\Rightarrow U = k + 3PV$

$\Rightarrow n = 3$

18. C

Sol. The force acting on the elementary portion of the current carrying conductor is given as,



$dF = I(dx)B\sin 90°$

$\Rightarrow dF = BIdx$

The torque applied by dF about O = $d\tau = xdF$

$\Rightarrow$ The total torque τ about O is given by

$\tau = \int d\tau = \int x(BIdx)$

$\Rightarrow \tau = IB\int_0^L xdx = \frac{IBL^2}{2}$

The angular acceleration $\alpha = \frac{\tau}{(\text{moment of inertia of rod about O})} = \frac{\tau}{\left(\frac{1}{3}mL^2\right)}$

$\Rightarrow \alpha = \frac{\left(\frac{BIL^2}{2}\right)}{\left(\frac{mL^2}{3}\right)} = \frac{3IB}{2m}$

19. B

Sol. Charge on smaller sphere

$q = C\left(\frac{V}{2}\right) = \frac{4\pi\varepsilon_0 RV}{2}$

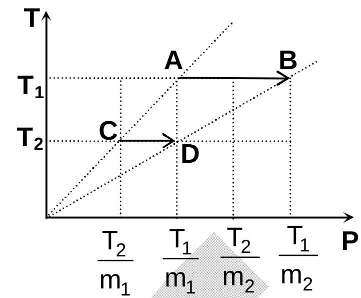
Potential difference = $\frac{q}{4\pi\varepsilon_0}\left(\frac{1}{R} - \frac{1}{2R}\right) = \left(\frac{VR}{2}\right)\left(\frac{1}{2R}\right) = \frac{V}{4}$

20. C

Sol. Let slope of line AC is m_1 and that of line DB is m_2, then

$$nRT_1\ell n\frac{m_2}{m_1} = 2nRT_2\ell n\frac{m_2}{m_1}$$

$$\frac{T_1}{T_2} = 2$$



SECTION – B

21. 16

Sol. $\ell = V_x t$, $V_x = \sqrt{\frac{2qV_0}{m}}$

$$V_y = \int \frac{qE}{m}dt$$

$$= \frac{q}{m}\int atdt$$

$$V_y = \frac{q}{m}a\frac{t^2}{2}$$

$$V_y = \frac{q}{m}a\frac{\ell^2}{2V_x^2}$$

$$= \frac{qa\ell^2}{2m}\left(\frac{m}{2qV_0}\right)$$

$$V_y = \frac{a\ell^2}{4V_0}$$

$$\tan\theta = \frac{V_y}{V_x} = \frac{a\ell^2}{4}\sqrt{\frac{m}{2qV_0^3}}$$

22. 4

Sol. $\phi_{ring} = MI_1$

$$\phi_{ring} = M(\alpha t^2)$$

$$\varepsilon = \frac{d\phi}{dt} = (M\alpha)(2t) = 2M\alpha t$$

23. 6

Sol. $\vec{\tau} = \vec{m}\times\vec{B} = 4\pi\times10^{-4}\times1500\pi\times10^{-6}$

$= 6 \times 10^{-6}$ Nm

24. 3

Sol. $U\rho$ = constant

$\frac{T}{V}$ = constant

$\Rightarrow$ P = constant

$$\frac{dU}{dW} = \frac{dU}{dQ - dU} = \frac{C_V}{C_P - C_V} = \frac{3}{2}$$

25. 18

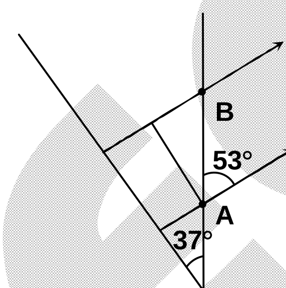
Sol. $t = nT = 3\left(\frac{2\pi m}{qB}\right) = \frac{6\pi m}{qB}$

$$y = \frac{1}{2}\left(\frac{qE}{m}\right)\left(\frac{36\pi^2 m^2}{q^2B^2}\right) = \frac{18\pi^2 E}{\alpha B^2}$$

26. 3

Sol. $E = \frac{\sigma}{2\varepsilon_0} = 1 \text{ N/C}$

$$\therefore\ V_B - V_A = -\int \vec{E}\cdot\overrightarrow{d\ell} = \left(\frac{\sigma}{2\varepsilon_0}\right)(5\cos 53^\circ) = 3V$$



27. 60

Sol. $Q = \frac{3}{4}mL$

2Q = mL + ms(50 –20)

$\Rightarrow\ \frac{3}{2}mL = mL + 30\text{ ms}$

$\Rightarrow\ \frac{L}{2} = 30\text{ s}$

$\Rightarrow\ \frac{L}{s} = 60$

28. 256

Sol. $P \propto T^4,\ T \propto \frac{1}{\lambda}$

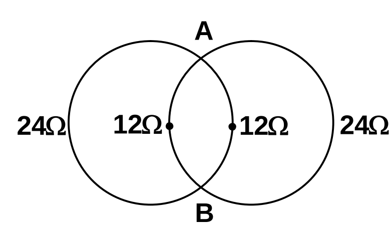
$$\therefore\ P \propto \frac{1}{\lambda_0^4}\ \Rightarrow P' = P\left(\frac{64\times 4}{81}\right)$$

29. 100

Sol. The resistance of four sections are shown in the figure.
Hence, the equivalent resistance R across AB is

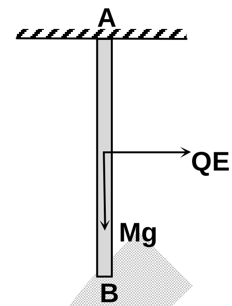
$\frac{1}{R} = \frac{1}{24} + \frac{1}{12} + \frac{1}{12} + \frac{1}{24}$ or R = 4Ω

Power = $\frac{V^2}{R} = \frac{(20)^2}{4} = 100\text{ W}$



30. 1

Sol. After turning through 90°, the rod comes at rest momentarily. So there is no change in K.E. Net work done by all the forces should be zero.



$$QE\frac{L}{2} - Mg\frac{L}{2} = 0$$

$$\Rightarrow E = \frac{Mg}{Q}$$

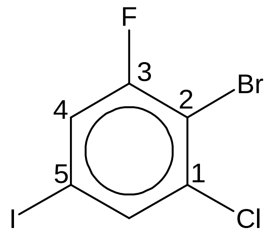
Chemistry

PART – B

SECTION – A

31. C

Sol.



2-Bromo-1-chloro-3-fluoro-5-iodobenzene

32. A

Sol.

$$Fe^{+3}(aq) + SCN^{-}(aq) \longrightarrow \left[Fe(SCN)(H_2O)_5\right]^{+2}$$

Blood red colour

So, the organic compound must contain S and N. So, option (A) is correct.

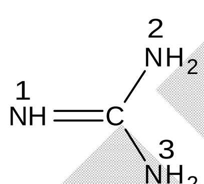
33. B

Sol.
1. Organic compound must contain – OH group as it shows CAN test.
2. It should also contain C = C as it decolorizes cold dilute alkaline $KMnO_4$.
3. It must contain $-\overset{\overset{\large OH}{|}}{CH}-CH_3$ group as it shows iodoform test.

So, option (B) is correct.

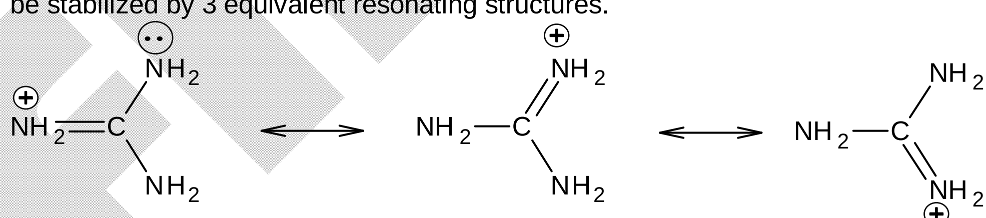
34. A

Sol.

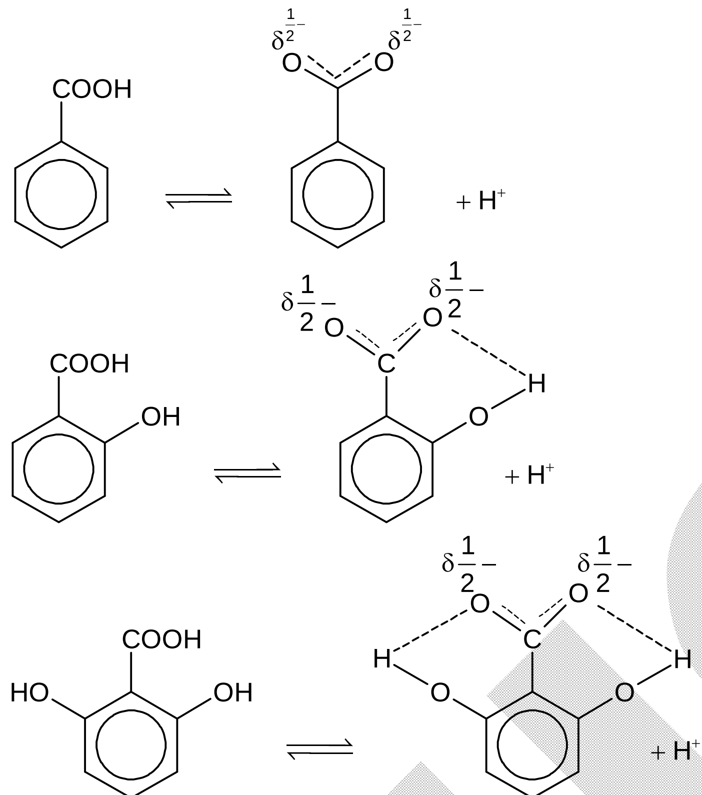


$\Longrightarrow$ Guanidine

In Guanidine, N – 1 should be protonated readily because in that case conjugate acid formed will be stabilized by 3 equivalent resonating structures.



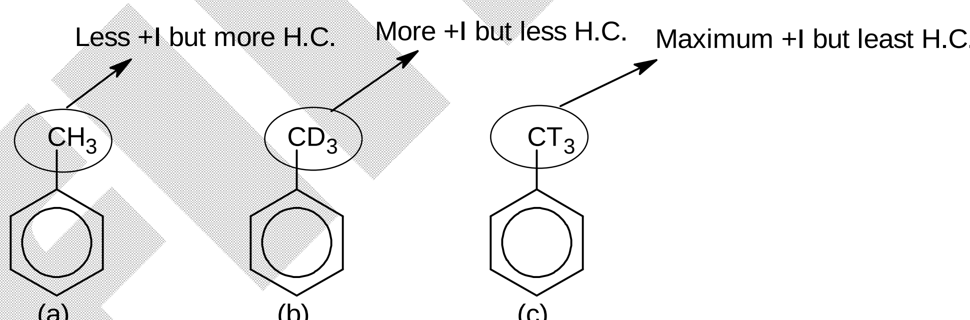
35. A
Sol.



So, benzoic acid will be least acidic and 2,6-dihydroxy benzoic acid will be most acidic. So. option (A) is correct.

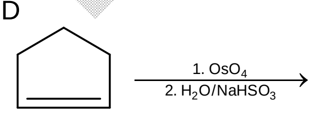
36. D
Sol. $\overset{\ominus}{CCl_3}$ is most stable among the anions given because it is stabilized by d-orbital resonance.

37. A
Sol.



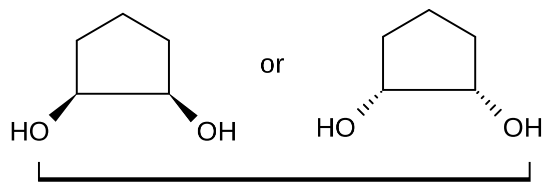
So, reactivity order towards Electrophilic aromatic substitution
a > b > c

38. D
Sol.



syn – hydroxylation takes place.

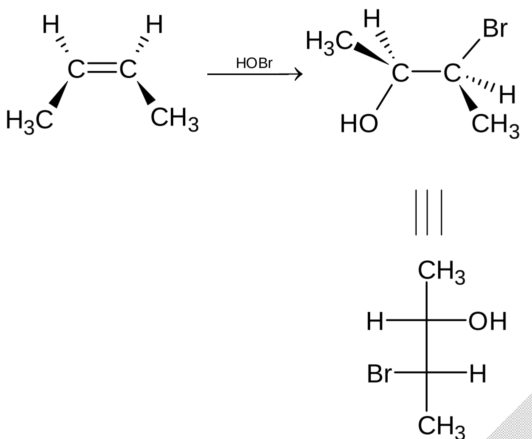
So, product will be



Meso product due to presence of POS

39. B

Sol.



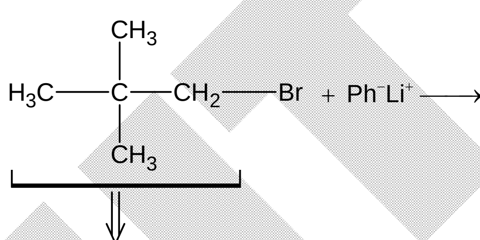
So, option (B) will be correct.

40. D

Sol. Number of geometrical isomers = $2^n = 2^3 = 8$

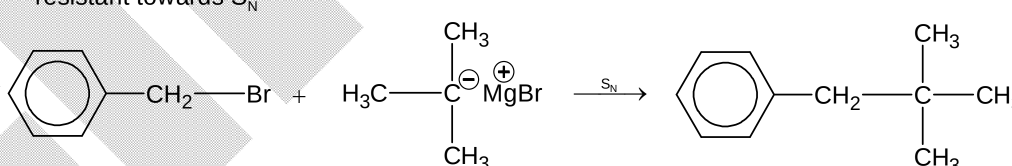
41. B

Sol. (A)



resistant towards S_N

(B)



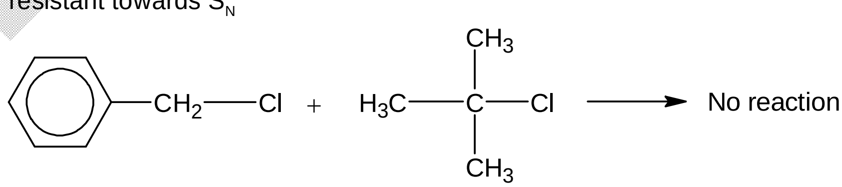
(C)

Neopentyl bromide + $Ph^{\ominus}\,MgBr^{\oplus} \longrightarrow$

$\Downarrow$

resistant towards S_N

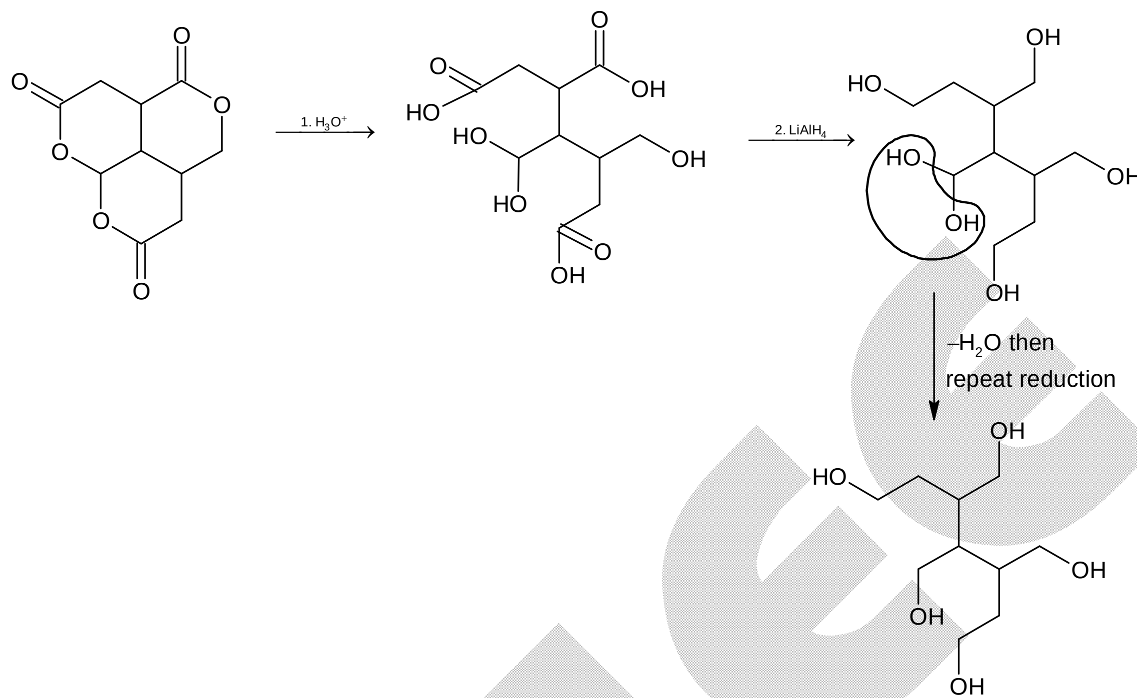
(D)



So, option (B) is correct.

42. B

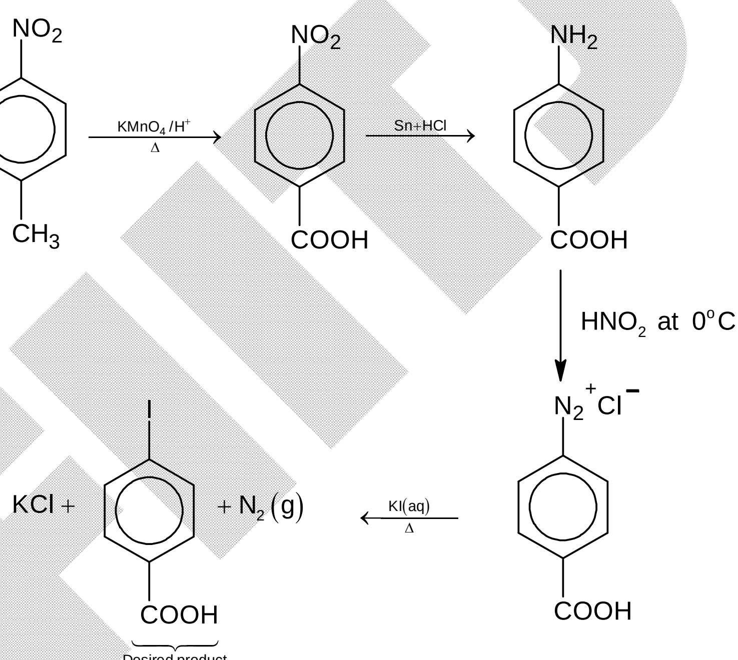
Sol.



So, option (B) is correct.

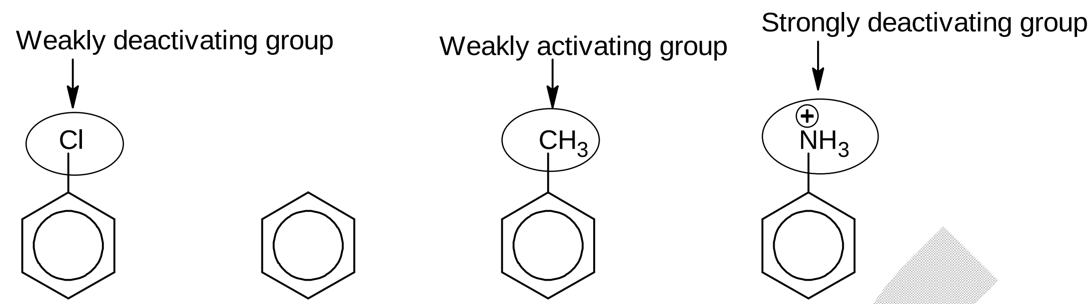
43. D

Sol.



So, option (D) is correct.

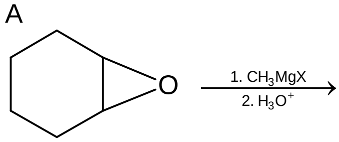
44. A
Sol.



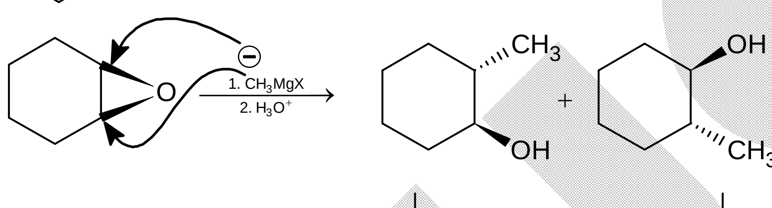
(I) (II) (III) (IV)

So, reactivity towards EAS will be
(III) > (II) > (I) > (IV)

45. A
Sol.

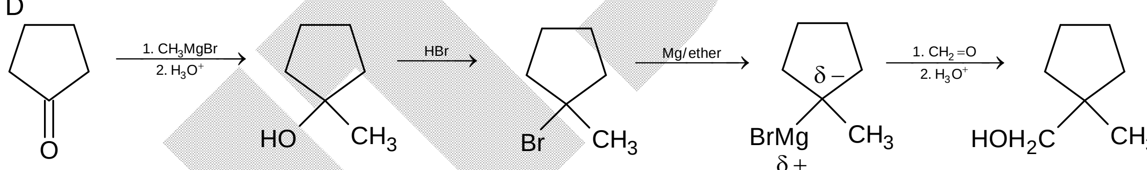


S_N2 reaction takes place



Mixture of enantiomers

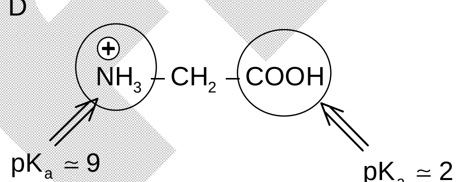
46. D
Sol.



So, option (D) will be correct.

47. C
Sol. Picric acid will react with $NaHCO_3$ to liberate effrevescence of $CO_2(g)$ but p-methoxy phenol won't.
So, option (C) is correct.

48. D
Sol.



(A) $\overset{\oplus}{N}H_3 - CH_2 - COOH$ is the dominating form at pH = 1. 0.

(B) $\overset{\oplus}{N}H_3 - CH_2 - \underset{\underset{O}{\|}}{C} - O^{\ominus}$ is the dominating form at pH = 6.0.

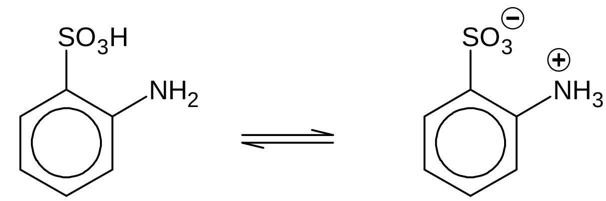
(C) $NH_2-CH_2-COO^-$ is the dominating form at pH = 10.0.

(D) $NH_2-CH_2-COO^-$ is the dominating form at pH = 12.0.

So, option (D) belongs to incorrect structure.

49. B

Sol.



Sulphanilic acid

In the Zwitter ion form of sulphanilic acid, the $SO_3^{\ominus}$ is too weakly basic (its conjugated acid $-SO_3H$ is a strong acid) to accept H^+ from HCl, hence it does not dissolve in HCl.

50. B

Sol. Factual

SECTION – B

51. 6

Sol. $-CH_3 \Rightarrow +I$ effect

$-\overset{\oplus}{N}H_3 \Rightarrow -I$ effect

$-OH \Rightarrow -I$ effect

$-O^- \Rightarrow +I$ effect

$-N(CH_3)_2 \Rightarrow -I$ effect

$-SO_3H \Rightarrow -I$ effect

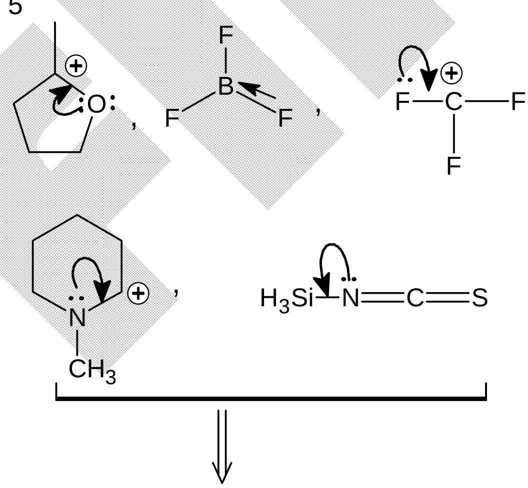
$-CHO \Rightarrow -I$ effect

$-Cl \Rightarrow -I$ effect

$-COO^- \Rightarrow +I$ effect

52. 5

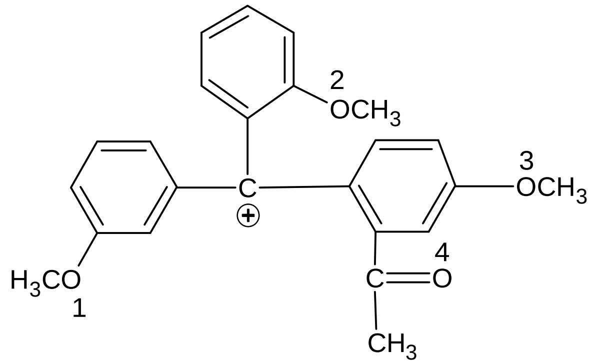
Sol.



All above examples possess back-bonding.

53. 2

Sol.



+ve charge of above carbocation will be delocalized over O – 2 and O – 3.

54. 5

Sol. $H_3C-Cl + CH_3-CH_2-Cl + CH_3-CH_2-CH_2-Cl$

$$\Big\downarrow \text{Dry ether} \;|\; \text{Na}$$

$$H_3C-CH_3 + CH_3-CH_2-CH_3 + CH_3-CH_2-CH_2-CH_3 + CH_3-CH_2-CH_2-CH_2-CH_3 + \text{Hexane}$$

55. 3

Sol.

$$\underset{\text{Br}\quad\ \text{Br}}{CH_3-CH-CH_2} \xrightarrow{\text{2 mole } NaNH_2} CH_3-C\equiv CH \xrightarrow{\text{1 mole } NaNH_2} CH_3-C\equiv \overset{\ominus}{C}\,Na^+$$

$$\Big\downarrow CH_3-CH_2-Br$$

$$CH_3-C\equiv C-CH_2-CH_3$$

So, total 3 moles of $NaNH_2$ are consumed to get pentyne.

56. 3

Sol. 2^o alcohols give blue colour during Victor-Meyer test.

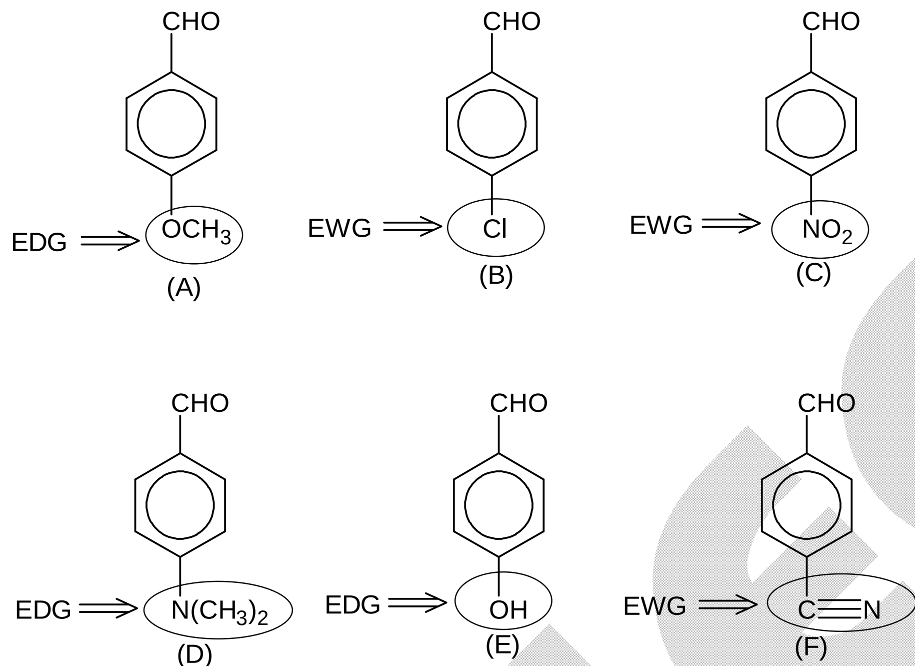
So, following 2^o pentyl alcohols are possible:

1. $CH_3-CH_2-CH_2-\underset{\large OH}{\overset{*}{C}H}-CH_3 \left< \begin{matrix} d \\ \ell \end{matrix} \right.$

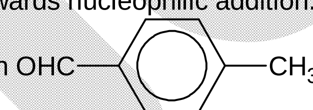
2. $CH_3-CH_2-\underset{\large OH}{CH}-CH_2-CH_3$

57. 3

Sol.

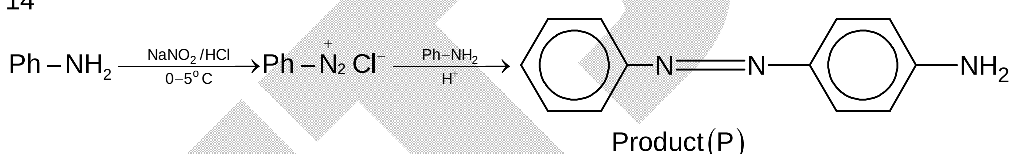


EWG groups increase reactivity of aldehyde towards nucleophilic addition.

So, molecules B, C, F will be more reactive than 

58. 14

Sol.



So,
X = 3, Y = 2, Z = 9
X + Y + Z = 14.

59. 6

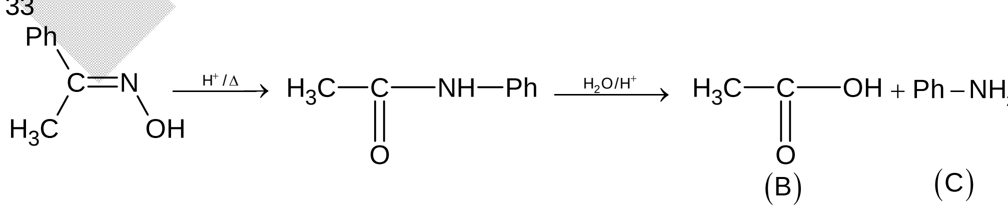
Sol. A decapeptide will require 9 molecules of H_2O as it contain 9 peptide bonds.
So, the total mass of the product obtained = 796 + 162 = 958

So, glycine by weight in the product will be $= 958 \times \frac{47}{100}$
$= 450.26$

So, number of glycine units in the decapeptide $= \frac{450.26}{75} = 6$

60. 33

Sol.



$|\text{Molar mass of B} - \text{Molar mass of C}| = |60 - 93| = 33$.

Mathematics

PART – C

SECTION – A

61. A

Sol. $y^2 = k^2x \Rightarrow y^2 = 4\left(\frac{k^2}{4}\right)x$

$B = \left(\frac{k^2}{4}t_1^2, \frac{k^2}{2}t_1\right), F = \left(\frac{k^2}{4}t_2^2, \frac{k^2}{2}t_2\right)$

$\frac{k^2}{4}t_1^2 = \frac{k^2}{2}t_1 \Rightarrow t_1 = 2$

$\frac{k^2}{2}t_2 = \frac{k^2}{4}\left(t_2^2 - t_1^2\right) \Rightarrow 2t_2 = t_2^2 - 4$

$\Rightarrow t_2^2 - 2t_2 - 4 = 0 \Rightarrow t_2 = 1+\sqrt{5}$

$\Rightarrow \frac{FG}{BC} = \frac{t_2}{t_1} = \frac{1+\sqrt{5}}{2}$

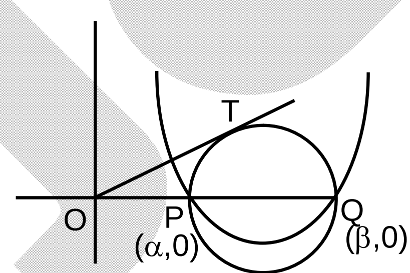
62. D

Sol. α, β are the roots of $ax^2 + bx + c = 0$

$\Rightarrow \alpha\beta = \frac{c}{a}$

$(OT)^2 = (OP)(OQ) = \alpha\beta = \frac{c}{a}$

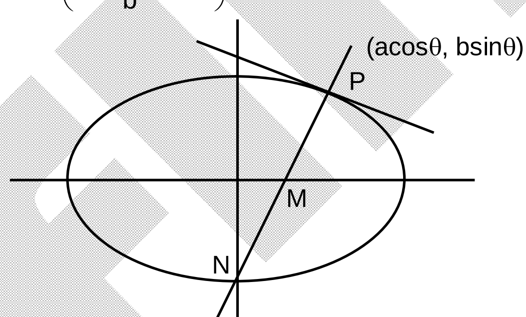
$OT = \sqrt{\frac{c}{a}}$



63. C

Sol. $M = (ae^2\cos\theta, 0)$

$N = \left(0, \frac{a^2e^2}{b}\sin\theta\right)$



$PM^2 = a^2\cos^2\theta(1 - e^2)^2 + b^2\sin^2\theta = b^2(1 - e^2\cos^2\theta)$

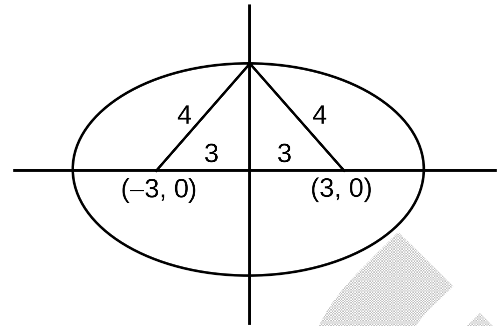
$PN^2 = a^2\cos^2\theta + \frac{\sin^2\theta}{b^2}(b^2 + a^2e^2)^2 = \frac{a^4}{b^2}(1 - e^2\cos^2\theta)$

$\left(\frac{PM}{PN}\right)^2 = \frac{b^4}{a^4} = \frac{4}{9} \Rightarrow \frac{b^2}{a^2} = \frac{2}{3} = 1 - e^2 \Rightarrow e = \frac{1}{\sqrt{3}}$

64. A

Sol. $a = 4$, $ae = 3$

$b^2 = 16 - 9 = 7$



65. A

Sol. Let the equation of line is $\frac{x}{a}+\frac{y}{b}=1$, it passes through (1, 1)

$\Rightarrow \frac{1}{a}+\frac{1}{b}=1$ also $h = \frac{a+b}{2}$, $k^2 = ab$

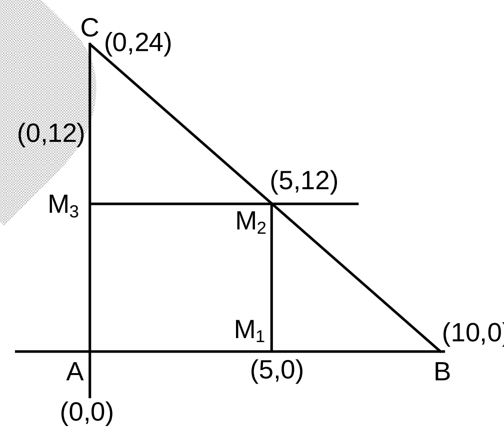
$\Rightarrow \frac{2h}{k^2}=\frac{a+b}{ab} \Rightarrow \frac{2h}{k^2}=\frac{1}{a}+\frac{1}{b}$

$= \frac{2h}{k^2}=1 \Rightarrow y^2 = 2x$

66. A

Sol. The vertices A, B, C will be (0, 0), (10, 0), (0, 24)

$\Rightarrow$ orthocentre (0, 0)



67. B

Sol. Put $x = -2y^2$ in line

$-2ay^2 + by + c = 0$

$y_1 + y_2 = \frac{b}{2a}=\frac{r}{2}$

68. C

Sol. Let P(x, y) be the point of contact $2y\frac{dy}{dx}=4a$ and $2x=4a\frac{dy}{dx}$. Now both $\frac{dy}{dx}$ are slope of same tangent.

$\Rightarrow \frac{4a}{2y}=\frac{2x}{4a} \Rightarrow xy = 4a^2$

69. B

Sol. Let ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$

P(acos2α, bsin2α) Q (acos2β, bsin2β) and R(acos2γ, bsin2γ)

chord PQ $\frac{x}{a}\cos(\alpha+\beta)+\frac{y}{b}\sin(\alpha+\beta)=\cos(\alpha-\beta)$

PQ passes through (ae, 0)

$\Rightarrow e=\frac{\cos(\alpha-\beta)}{\cos(\alpha+\beta)}$

Similarly PR passes through (–ae, 0)

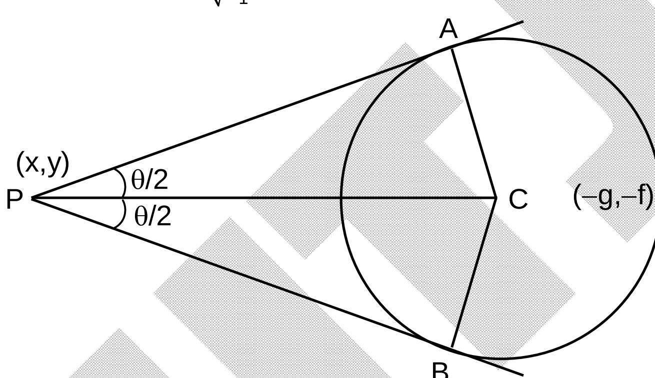
$\Rightarrow -e=\frac{\cos(\alpha-\gamma)}{\cos(\alpha+\gamma)}\Rightarrow\frac{\cos(\alpha-\beta)}{\cos(\alpha+\beta)}=-\frac{\cos(\alpha-\gamma)}{\cos(\alpha+\gamma)}$

$\Rightarrow \frac{\cos(\alpha+\beta)+\cos(\alpha-\beta)}{\cos(\alpha+\beta)-\cos(\alpha-\beta)}=\frac{\cos(\alpha+\gamma)-\cos(\alpha-\gamma)}{\cos(\alpha+\gamma)+\cos(\alpha-\gamma)}$

$\Rightarrow \frac{2\cos\alpha\cos\beta}{2\sin\alpha\sin\beta}=\frac{2\sin\alpha\sin\gamma}{2\cos\alpha\cos\gamma}\Rightarrow \tan\beta\tan\gamma=\cot^2\alpha$

70. A

Sol. $\tan\frac{\theta}{2}=\frac{CA}{PA}=\frac{\sqrt{g^2+f^2-c}}{\sqrt{s_1}}$



$\theta=2\tan^{1}=\sqrt{\frac{g^2+f^2-c}{s_1}}\Rightarrow$ (C)

$\Rightarrow \cot\frac{\theta}{2}=\frac{\sqrt{s_1}}{\sqrt{g^2+f^2-c}}\Rightarrow$ (B)

and $\cos\theta=\frac{1-\tan^2\frac{\theta}{2}}{1+\tan^2\frac{\theta}{2}}=\frac{s_1-(g^2+f^2-c)}{s_1+g^2+f^2-c}\Rightarrow$ (D)

$\Rightarrow$ (A) is incorrect.

71. C

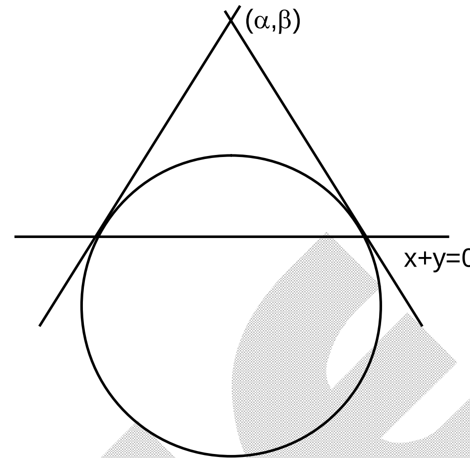
Sol. Equation of chord of contact through (α, β) is

$x\alpha + y\beta + k(x + \alpha) + k(y + \beta) - 8 = 0$

$x(\alpha + k) + y(\beta + k) + k\alpha + k\beta - 8 = 0$

Compare with $x + y = 0$

$\alpha + k = \beta + k \Rightarrow \alpha = \beta \Rightarrow x = y$ is locus.



72. D

Sol. The point of intersection is $\left(\frac{ab}{a+b}, \frac{ab}{a+b}\right)$

73. B

Sol. Let $P(a\sec\theta, b\tan\theta)$ and $Q(a\tan\phi, b\sec\phi)$ since P and Q lie on the line parallel to $x = 0$

$\Rightarrow a\sec\theta = a\tan\phi$

Now normal at P is $y - b\tan\theta = -\frac{a\tan\theta}{b\sec\theta}(x - a\sec\theta)$

and normal at Q is $y - b\sec\phi = -\frac{a\sec\phi}{b\tan\phi}(x - a\tan\phi)$

use $\sec\theta = \tan\phi$ we get

$\frac{b\sec\theta}{a\tan\theta}(y - b\tan\theta) = \frac{b\tan\phi}{a\sec\phi}(y - b\sec\phi) \Rightarrow y = 0$

74. A

Sol. Now $2\theta = \frac{5}{r} \Rightarrow \theta = \frac{5}{2r}$

$\frac{OP}{\sin(180-2\theta)} = \frac{CP}{\sin\theta} \Rightarrow \frac{\sqrt{3}h}{2\sin\theta\cos\theta} = \frac{r}{\sin\theta}$

$\Rightarrow \cos\theta = \frac{\sqrt{3}h}{2r}$

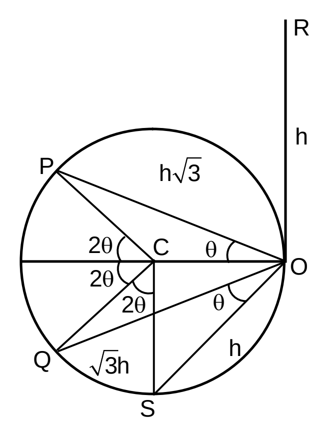
again from Δcos $\frac{OS}{\sin(180-4\theta)} = \frac{CS}{\sin 2\theta} \Rightarrow \cos 2\theta = \frac{h}{2r}$

$\Rightarrow 2\cos^2\theta - 1 = \frac{h}{2r} \Rightarrow \frac{2(3h^2)}{4r^2} - \frac{h}{2r} - 1 = 0 \Rightarrow \frac{h}{r} = 1$

$\Rightarrow \cos\theta = \frac{\sqrt{3}}{2} \Rightarrow \theta = \frac{\pi}{6}$

Also $2\theta r = 5 \Rightarrow r = \frac{5}{2\theta} = \frac{15}{\pi}$

Perimeter $= 2\pi r = 30$ meters.



75. B

Sol. $3\sin\alpha - 4\sin^3\alpha = 4\sin\alpha(\sin^2 x - \sin^2\alpha)$

$\Rightarrow \sin^2 x = \frac{3}{4} \Rightarrow \sin x = \frac{\sqrt{3}}{2}$

$\Rightarrow$ 2 solutions $x = \frac{\pi}{3}, \frac{2\pi}{3}$

76. A

Sol. Let $\frac{\cos\theta}{a} = \frac{\sin\theta}{b} = k \Rightarrow \cos\theta = ak$, $\sin\theta = bk$, then $a\cos2\theta + b\sin2\theta$

$= a(1 - 2\sin^2\theta) + b(2\sin\theta\cos\theta)$

$= a - 2ab^2k^2 - b(2bk)(ak) = a$

77. C

Sol. $\sin(\alpha - \beta) = \sin(\alpha - \theta + \theta - \beta) = \sin((\theta - \beta) - (\theta - \alpha))$

$= \sin(\theta - \beta)\cos(\theta - \alpha) - \cos(\theta - \beta)\sin(\theta - \alpha)$

$= ba - \sqrt{1-b^2}\sqrt{1-a^2}$

$\cos(\alpha - \beta) = a\sqrt{1-b^2} + b\sqrt{1-a^2}$

$\cos^2(\alpha - \beta) + 2ab\sin(\alpha - \beta) = a^2 + b^2$

78. C

Sol. Put $x = \cos\theta$, $0 \le \theta \le \frac{\pi}{3}$

$\cos^{-1}(\cos\theta) + \cos^{-1}\left(\frac{1}{2}\cos\theta + \frac{\sqrt{3}}{2}\sin\theta\right)$

$= \theta + \cos^{-1}\cos\left(\frac{\pi}{3} - \theta\right)$

$= \theta + \frac{\pi}{3} - \theta = \frac{\pi}{3}$

79. C

Sol. Let $\tan^{-1}\sqrt{\cos\alpha} = \theta \Rightarrow \tan\theta = \sqrt{\cos\alpha} \Rightarrow \cos\alpha = \tan^2\theta$

$x = \frac{\pi}{2} - \tan^{-1}\sqrt{\cos\alpha} - \tan^{-1}\sqrt{\cos\alpha}$

$\Rightarrow x = \frac{\pi}{2} - 2\theta \Rightarrow \sin x = \cos 2\theta = \frac{1-\tan^2\theta}{1+\tan^2\theta}$

$\Rightarrow \sin x = \frac{1-\cos\alpha}{1+\cos\alpha} = \tan^2\frac{\alpha}{2}$

80. A

Sol. $\cot\frac{A}{2} + \cot\frac{B}{2} + \cot\frac{C}{2} = \sqrt{\frac{s(s-a)}{(s-b)(s-c)}} + \sqrt{\frac{s(s-a)}{(s-b)(s-c)}} + \sqrt{\frac{s(s-c)}{(s-a)(s-b)}}$

$= \frac{\sqrt{s}\sqrt{s}(s-a+s-b+s-c)}{\sqrt{s}\sqrt{(s-a)(s-b)(s-c)}} = \frac{s^2}{\Delta} = \frac{(a+b+c)^2}{4\Delta}$

$\cot A + \cot B + \cot C = \frac{\cos A}{\sin A} + \frac{\cos B}{\sin B} + \frac{\cos C}{\sin C}$

$$= \frac{b^2+c^2-a^2}{2bc\sin A}+\frac{c^2+a^2-b^2}{2ca\sin B}+\frac{a^2+b^2+c^2}{2ab\sin C} = \frac{a^2+b^2+c^2}{4\Delta}$$

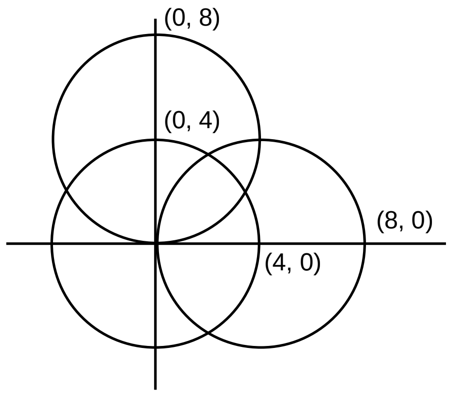
SECTION – B

81. 6

Sol. Let $S_1 = x_1^2+y_1^2-16$

$S_2 = x_1^2+y_1^2-8x$

$S_3 = x_1^2+y_1^2-8y$



If (x_1, y_1) lies inside the three circles then $S_1 < 0$, $S_2 < 0$, $S_3 < 0$.
Now (1, 1), (2, 2), (1, 2), (2, 1), (2, 3), (3, 2) are the points satisfying all the three in equations.

82. 7

Sol. Equation of transverse common tangent with positive slope is $y - 5 = 2\sqrt{2}(x-1)$

$\Rightarrow 2\sqrt{2}x - y + (5-2\sqrt{2}) = 0$

Now line parallel to this and tangent to C_2 can be $2\sqrt{2}x - y = \lambda$

$\left|\frac{2\sqrt{2}(1)-8-\lambda}{\sqrt{8+1}}\right| = 1$

$\Rightarrow 2\sqrt{2}-8-\lambda = \pm 3$

$\lambda = 2\sqrt{2}-11,\ 2\sqrt{2}-5$

Now equation of L_2 is $2\sqrt{2}x - y + 11 - 2\sqrt{2} = 0$

$\frac{a+b+c}{2} = \frac{2+1+11}{2} = 7$

83. 2

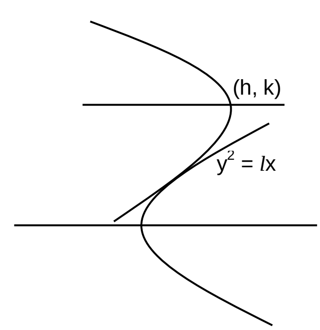
Sol. Let (h, k) is the vertex of the variable parabola.
Now equation of the variable parabola is $(y-k)^2 = l(h-x)$

$\Rightarrow 2y^2 - 2ky + k^2 - lh = 0$

$D = 0$

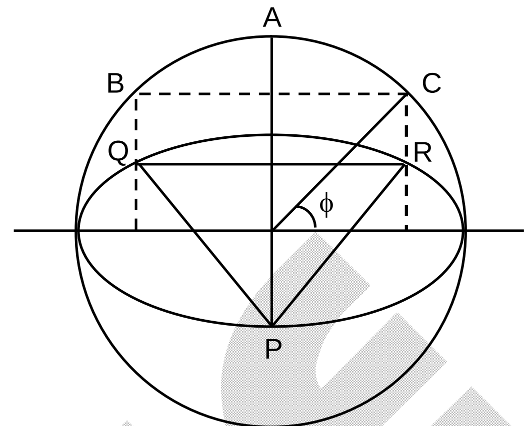
$\Rightarrow 4k^2 - 8(k^2 - hl) = 0$

$\Rightarrow k^2 = 2lh \Rightarrow y^2 = 2lx$



84. 24

Sol. Given A = (0, 4)
Q = (–4cosϕ, 3sinϕ)
R = (4cosϕ, 3sinϕ)
P = (0, –4)



$PQ \perp QR \Rightarrow \left(\frac{3(\sin\phi+1)}{4\cos\phi}\right)\left(\frac{3(\sin\phi+1)}{-4\cos\phi}\right) = -1$

$\Rightarrow 25\sin^2\phi + 18\sin\phi + 7 = 0 \Rightarrow \sin\phi = \frac{7}{25}$

$\Rightarrow \tan\frac{\phi}{2} = \frac{1}{7} \Rightarrow \theta = \angle ABC = \frac{1}{2}\left(\frac{\pi}{2} - \phi\right)$

$\tan\theta = \frac{1-\tan(\phi/2)}{1+\tan(\phi/2)} = \frac{3}{4} \Rightarrow 16\tan^2\theta + 8\tan\theta + 9 = 24$

85. 4

Sol. Let the point of contact of tangent is (acosθ, bsinθ) how equation of tangent $\frac{x}{a}\cos\theta + \frac{y}{b}\sin\theta = 1$. It is normal to circle hence passes through the centre (–2, 0)

$\Rightarrow -\frac{2}{a}\cos\theta = 1 \Rightarrow \sec\theta = -\frac{2}{a}$

Slope of tangent = $-\frac{b}{a}\cot\theta = 2$

$\Rightarrow \tan\theta = -\frac{b}{2a} \Rightarrow 1 + \frac{b^2}{4a^2} = \frac{4}{a^2}$

$\Rightarrow 16 = 4a^2 + b^2 \geq 2(2ab) \Rightarrow \max(ab) = 4$

86. 2

Sol. Let y = mx + c is common tangent

$\Rightarrow mx + c = x^2 - x + 1 \Rightarrow x^2 - (m + 1)x + 1 - c = 0$

$\Rightarrow (m + 1)^2 - 4(1 - c) = 0$ also $mx + c = x^2 - 3x + 1$

$\Rightarrow x^2 - (m + 3)^2 + 1 - c = 0 \Rightarrow (m + 3)^2 - 4(1 - c) = 0$

$\Rightarrow (m + 1)^2 = (m + 3)^2 \Rightarrow m = -2$

87. 3

Sol. $b^2 + c^2 = 2\left(\alpha^2 + \frac{a^2}{4}\right) \Rightarrow b^2 + c^2 = 2\alpha^2 + \frac{a^2}{2}$

$\Rightarrow 2b^2 + 2c^2 - a^2 = 4\alpha^2$

$2c^2 + 2a^2 - b^2 = 4\beta^2$

$2a^2 + 2b^2 - c^2 = 4\gamma^2$

$\Rightarrow 3(a^2 + b^2 + c^2) = 4(\alpha^2 + \beta^2 + \gamma^2)$

88. 8

Sol. Simplify get sin2x = cos2x ⇒ tan2x = 1

$\Rightarrow 2x = n\pi + \frac{\pi}{4} \Rightarrow x = n\pi + \frac{\pi}{8}$

89. 1

Sol. Put $\theta = \frac{\pi}{2}$ $\Rightarrow 1 = a + b + c + d$

90. 6

Sol. $\frac{\sin\theta}{\cos 3\theta} = \frac{2\sin\theta\cos\theta}{2\cos 3\theta\cos\theta} = \frac{\sin 2\theta}{2\cos\theta\cos 3\theta}$

$= \frac{\sin(3\theta - \theta)}{2\cos\theta\cos 3\theta} = \frac{1}{2}\left(\frac{\sin 3\theta\cos\theta}{\cos 3\theta\cos\theta} - \frac{\cos 3\theta\sin\theta}{\cos 3\theta\cos\theta}\right)$

$= \frac{1}{2}$ (tan3θ – tanθ) $\Rightarrow$ the equation be comes

tan27x = tanx $\Rightarrow$ 27x = nπ = x $\Rightarrow$ x = $\frac{n\pi}{26}$

Possible n = 1, 2, 3, 4, 5, 6